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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE GUIDE

PART 6

CHARACTERIZATION OF THE ELECTROMAGNETIC ENVIRONMENT FOR HERO

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This part supplements MIL-HDBK-240-1 and was prepared for contractors, DoD test agencies, activities, and test planners. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides recommended practices for evaluating the electromagnetic environment at facilities or platforms where ordnance with electrically initiated devices (EIDs) are utilized, stored, and/or handled. Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, and so forth, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commanders Group (JOCG) Electromagnetic Environmental Effects (E3) Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

5. Comments, recommendations, additions, or deletions and any other pertinent data that may improve this document should be emailed to marcus.shellman.civ@mail.mil or addressed to:

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 supplements DESR 6055.09 by providing the guidance necessary to characterize the electromagnetic environment (EME) to mitigate hazards of electromagnetic radiation to ordnance (HERO).

Technological advances have resulted in the continued proliferation of communication and radar equipment, many of which radiate high levels of electromagnetic (EM) energy. Advances in weapon technology have increased the use of sensitive, low-power electronic circuits in the design of ordnance systems, presenting the possibility of unintended actuation of or damage to the circuits caused by this EM energy. This concern is addressed by conducting comprehensive radiation hazard assessments through mathematical modeling and analysis, on-site measurements and evaluations, or a combination of methods. These assessments characterize the level of radio frequency (RF) EMEs so that guidance can be provided to minimize safety and reliability hazards while maintaining mission operability in the RF environment. This process is termed HERO assessment.

The results from the various HERO assessments can be used to update MIL-HDBK-235 and ultimately the HERO Test EME found in MIL-STD-464. In addition, the data is contributing to North Atlantic Treaty Organization's (NATO's) efforts to capture their operational EME.

1.2 Background. There are four specific objectives of this handbook:

a. Document the Joint-Service process to characterize the EME.

b. Promote standardization.

c. Identify alternative techniques and instrumentation.

d. Facilitate the exchange of data.

Ultimately, the information gathered is used to assist in developing the operational guidance (Part 7 of this document) for ordnance and weapon systems containing electrically initiated devices (EIDs).

When defining the operational EME, one should anticipate not only the individual Service EME, but also the EME in Joint-operation scenarios. JROCM 102-05 states that "Because all weapons/weapon systems have the potential of being deployed together or employed in joint environments, weapons and weapon systems will be considered joint systems within the Joint Capabilities Integration and Development System (JCIDS) process unless they are assigned the Joint Potential Designator of 'Independent.'" As Joint operations become more commonplace, there is an increased likelihood that weapons from one Service will be exposed to other Services' EME levels.

2. APPLICABLE DOCUMENTS

2.1 General. See MIL-HDBK-240-1. This document can be found on ASSIST Online at https://assist.dla.mil.

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3. DEFINITIONS

3.1 Definitions. See MIL-HDBK-240-1.

4. HERO ASSESSMENTS. From a HERO perspective there are a number of types of assessments (analytical and on-site) used to characterize the EME such that HERO can be mitigated:

a. HERO analyses.

1) Far-field EME calculations.

2) Near-field EME calculations.

b. HERO survey measurements.

1) Open-air measurements.

2) Complex cavity measurements (walk-about techniques).

c. Platform system surveys (vehicles and aircraft).

4.1 HERO Analyses. In many instances, it is not possible to conduct an on-site survey (measurements) for a specific platform/installation. Through the use of equations and Joint HERO Curves, one can calculate the safe separation distances (SSDs) for each emitter system and then determine if those arcs encompass ordnance locations. The use of this calculation methodology provides a relatively accurate determination of the maximum field strength from the transmitter sites and the SSD requirements. This approach is used mostly at installations but may be used for platform transmitters; however, the results are subject to beam reflections that may cause severe differences from those calculated. The only accurate gauge of power density/field intensity is actual measurement of the EME. This approach also assumes main-beam illumination and is more accurate for line-of-sight (LOS) applications.

4.1.1 Peak and Average Power Calculations. Typically, communication systems are capable of modulation techniques, such as amplitude modulation (AM), frequency modulation (FM), pulse-code modulation (PCM), or continuous wave (CW). For AM, the peak envelope power (PEP) of a 100 percent modulated AM signal is four times the carrier peak power and, therefore, is used to provide the worst-case scenario from AM signals. For CW, the duty cycle of a CW signal equals unity and the average and peak root mean square (rms) power are the same. For FM and PCM, in order to determine the rms peak power, a worst-case approach is used where the peak power is equal to the unmodulated carrier peak power. This rationale is used when calculating EMEs because of the randomly changing nature of true peak power over a specific interval. Pulse-modulated signals, typically radars, have differences between peak and average rms power. The average power is determined by the ratio of time-on to time-off over an interval. This time-on/off ratio is the duty cycle (DC) and can be calculated using equation (4-1). The average power can be calculated by the product of peak power and duty cycle as shown in equation (4-2).

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$$DC = \frac{pw}{pri} = pw \times prf \quad (4\text{-}1)$$

$$P_{t\,avg} = P_{t\,peak} \times DC \quad (4\text{-}2)$$

where:

DC is the duty cycle (unitless)
Pw is the pulse width [seconds (sec)]
pri is the pulse repetition rate interval (sec)
prf is the pulse repetition rate frequency [hertz (Hz)]
$P_{t\,avg}$ is the average power transmit [watts (W)]
$P_{t\,peak}$ is the peak power transmit (W)

4.1.2 Antenna Field Regions. The fields around an antenna are divided into three regions: the reactive near-field; the radiating near-field or Fresnel; and the far-field or Fraunhofer. The boundaries for these field regions are defined by equations (4-3) through (4-6), respectively.

$$NF_r \leq 0.62 \times \sqrt{\frac{D^3}{\lambda}} \quad (4\text{-}3)$$

$$0.62 \times \sqrt{\frac{D^3}{\lambda}} > NF_{rad} < \frac{2 \times D^2}{\lambda} \text{ for aperture antennas} \quad (4\text{-}4)$$

or

$$0.62 \times \sqrt{\frac{D^3}{\lambda}} > NF_{rad} < \frac{\lambda}{2\pi} \text{ for wire antennas} \quad (4\text{-}5)$$

$$d_{ff} \geq \frac{2 \times D^2}{\lambda} \quad (4\text{-}6)$$

where:

NF_r is the reactive near-field region [meters (m)]
D is the largest dimension of antenna (m)
λ is the wavelength (m)
NF_{rad} is the radiating near-field region (m)
d_{ff} is the far-field region (m)

Note that if the antenna is small compared to the wavelength (λ > 10D), the radiating near-field does not exist.

4.1.3 Power Density and Field Strength Calculations. The HERO SSDs are all derived using a far-field equation. In the far-field region, the power density is calculated using equation (4-7). All power density levels are calculated within the 3 decibel (dB) beam width of the main beam.

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$$P_D = \frac{P_t \times G_t}{4 \times \pi \times d^2} \quad (4\text{-}7)$$

where:

P_D is the power density [watts per square meter (W/m^2)]
P_t is the average or peak transmitter output power (W)
G_t is the numerical antenna gain (unitless)
d is the distance or range from the antenna (m)

In the near-field region, the power densities are calculated using the far-field equation (4-7) and a near-field gain reduction factor (N) equation (4-9). See 4.1.5.2.2, for calculation of the near-field gain reduction factor. Power density is related to the electric field (E-field) by equation (4-8). When converting from power density to field strength (or vice versa), the following relationship only exists for a plane wave (within the far field of the antenna where the relationship between the E-field and magnetic field (H-field) is orthogonal and clearly defined).

$$E = \sqrt{P_D \times Z_0} \quad (4\text{-}8)$$

where:

E is the maximum E-field strength [volts per meter-root-mean-square (V/m-rms)]
Z_0 is the intrinsic impedance of free space [ohms (Ω)]
P_D is the power density (W/m^2)

In the near-field region, an antenna's E- and H-fields do not exhibit a constant ratio of 120π (approximately 377 Ω), the intrinsic impedance of free space. Depending on the source antenna's terminal voltage, impedance, and driver current, the E- and H-fields will vary at different rates where one field becomes dominant. As the far-field region is approached, the ratio of the E- and H-fields begins to approximate 377 Ω; variation between the fields becomes less, and any dominance of one field are reduced.

4.1.4 Far-Field Power Density Calculations. The following paragraphs illustrate the use of the Joint HERO curves (MIL-HDBK-240-7) and how a maximum allowable environment (MAE) (which is frequency-dependent) is applied to the Friis Free-Space Equation to calculate the HERO SSD for ordnance classified as either HERO SUSCEPTIBLE or HERO UNSAFE. The ensuing paragraphs also provide the limitations of this approach. For example, when calculating an SSD, it is important to use average power (and not peak power) as the input parameter for calculating an SSD. It is the average power that most often affects EIDs; however, as the pulse width approaches one tenth of the thermal time constant or response time, then the peak power should be considered. For the HERO curves and the associated equations, it is important to understand that the results given are only accurate in the far field (see 4.1.3) and, consequently, discussions related to the near-field regions associated with an antenna require a different calculation methodology (see 4.1.5).

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4.1.5 Near-Field Power Density Calculation Methodology.

4.1.5.1 Introduction. More often than not, the far-field equations and the approach for calculating SSDs are adequate to address HERO. In certain instances, such as when ordnance is in proximity to high power antennas, a more precise approach is needed to calculate the EME and required SSD to mitigate HERO.

The Joint Ordnance Commanders Group (JOCG) Electromagnetic Environmental Effects (E3) Ordnance Safety Subgroup (formerly JOCG HERO Subgroup) developed a computer-based software program (MAE Analysis Program, available from the Joint Spectrum Center) capable of predicting the theoretical maximum response of an ordnance system's EIDs to a wide range of EMEs and translating this information into Service guidance in the form of MAEs. In addition, the program is also capable of calculating the distance from a given emitter source at which an ordnance system will remain safe and reliable. The calculations are based on the characteristics of the transmitter/antenna system and the ordnance system's MAE for the frequency range of concern. The SSD calculations consider near-field as well as far-field effects.

4.1.5.2 Near-Field Power Density Calculation. The method used to calculate the power density along the propagation axis of a large-aperture circular or rectangular antenna source in the "radiating" near-field region (Fresnel region) is provided in 4.1.5.2.1 and 4.1.5.2.2, respectively. Antenna gain and beam width both are degraded in the Fresnel region; therefore, the far-field power density equation (4-7) is modified to account for the antenna near-field factor (N). (See equation (4-9).)

$$P_D = \frac{P_t \times G_t}{4 \times \pi \times d^2} \times N \qquad (4\text{-}9)$$

where:

P_D is the average or peak power density (W/m^2)
P_t is the average or peak transmitter output power (W)
G_t is the numerical antenna gain (unitless)
N is the antenna near-field factor
d is the distance or range from the antenna (m)

The near-field power density calculation method used in calculating the EME levels is based on a consistent approach for estimating the antenna illumination and calculating/verifying the antenna efficiency. It does not consider the following: mismatch [voltage standing wave ratio (VSWR)] loss derived from the reflection at the antenna feed port because of impedance mismatch, RF losses between the antenna and the antenna feed point, and spillover loss and phase error loss resulting from the fact that the antenna aperture is not a uniform phase surface. The method for calculating the power density for a circular aperture in the near field is provided in 4.1.5.2.1, and the method for calculating the power density for a rectangular aperture in the near field is provided in 4.1.5.2.2. The two methods are slightly different in that the rectangular aperture method computes a gain reduction value for the antenna, whereas the circular method computes a near-field correction factor that includes both a gain reduction value and the space loss value.

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4.1.5.2.1 Near-Field Power Density Calculation Method for Circular-Aperture Antenna. The equation used to calculate the power density along the propagation axis of a large-aperture circular antenna source in the "radiating" near-field region is provided in equation (4-10). The antenna gain and beam width both are degraded in the Fresnel region; therefore, the far-field power density equation (4-7) is modified to account for the antenna near-field correction factor (NCF_{circ}).

$$P_D = \frac{P_t \times G_t}{4 \times \pi \times d_{ff}^2} \times NCF_{circ} \qquad (4\text{-}10)$$

where:

P_D is the average or peak power density (W/m^2)
P_t is the average or peak transmitter output power (W)
G_t is the numerical antenna gain (unitless)
NCF_{circ} is the near-field correction factor (unitless)
d_{ff} is the distance from the antenna to the far-field using equation (4-6) (m)

Steps for computing the near-field power density:

a. Compute the power density at the far-field boundary using the Friis transmission formula (4-7).

b. Calculate the illumination constant (R) using equation (4-11).

c. Estimate the antenna illumination and antenna factor using the illumination constant and TABLE I.

d. Check to see if the antenna illumination is reasonable by computing the antenna efficiency (K) using equation (4-12).

e. Calculate the normalized distance from the antenna and then determine the *NCF* using the appropriate antenna illumination in FIGURE 1.

f. Multiply the power density (at far-field boundary) computed in step a by the *NCF* determined in step e.

As stated above, the near-field correction factor depends on the type of antenna illumination and the distance from the antenna. If the antenna illumination is not known, it can be estimated by first calculating the illumination constant (R) using equation (4-11).

$$R = 5.817 \times 10^{-5} \times f \times BW \times D \qquad (4\text{-}11)$$

where:

R is the constant for estimating illumination
f is the frequency in megahertz (MHz)
BW is the beam width in degrees (horizontal or vertical at 3 dB points)

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D is the diameter of circular antenna (or largest horizontal or vertical dimension of rectangular antenna) (m)

After calculating R, the antenna illumination can be estimated from TABLE I. Illuminations above $(1-r^2)^4$ are omitted purposely since the gain reduction in the Fresnel region would be almost negligible.

TABLE I. Circular-Aperture Antenna Illuminations $(1-r^2)^\rho$

Limits of R	Estimated Illumination	Illumination (ρ)	Antenna Illumination Factor (F)
1.02 to 1.27	Uniform	0	1.00
1.27 to 1.47	$(1-r^2)$ Taper	1	0.75
1.47 to 1.65	$(1-r^2)^2$ Taper	2	0.56
1.65 to 1.81	$(1-r^2)^3$ Taper	3	0.44
> 1.81	$(1-r^2)^4$ Taper	4	0.36

When the illumination constant (R) is determined to be borderline between two orders of illumination (ρ), the higher order is selected because it will produce the maximum field strength. However, the antenna efficiency should be checked using equation (4-12) to determine if it is reasonable for this antenna type and illumination. Should the higher-order illumination cause the antenna efficiency to be too high, then the next lower order should be selected. Antenna efficiency (K) between 0.3 and 0.9 is considered reasonable.

$$K = \frac{G_t \times \lambda^2}{4\pi \times A \times F} \qquad (4\text{-}12)$$

where:

K is the antenna efficiency (unitless)
G_t is the far-field antenna gain (unitless)
A is the antenna physical area (m^2)
λ is the wavelength (m)
F is the antenna illumination factor (derived from TABLE I)

Once the antenna illumination factor (F) has been determined and the efficiency verified to be reasonable, then the appropriate gain correction factor can be applied to calculations in the near field for a circular antenna of a specific illumination type.

Now that the illumination type (ρ = 0, 1, 2, 3, 4) has been determined, the near-field correction factor can be selected by calculating the normalized distance from the antenna using equation (4-13) and then selecting the appropriate correction factor (NCF_{circ}) based on the illumination type from FIGURE 1.

$$x = \frac{d}{2 \times (D^2/\lambda)} \qquad (4\text{-}13)$$

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where:

x is the normalized distance from antenna
d is the distance from antenna
D is the diameter of circular antenna (or largest horizontal or vertical dimension of rectangular antenna) in meters
λ is the wavelength (m)

The near-field power density level at given distance (d) from the antenna can be determined by computing the power density at the far-field distance and multiplying this value by the correction factor (NCF_{circ}).

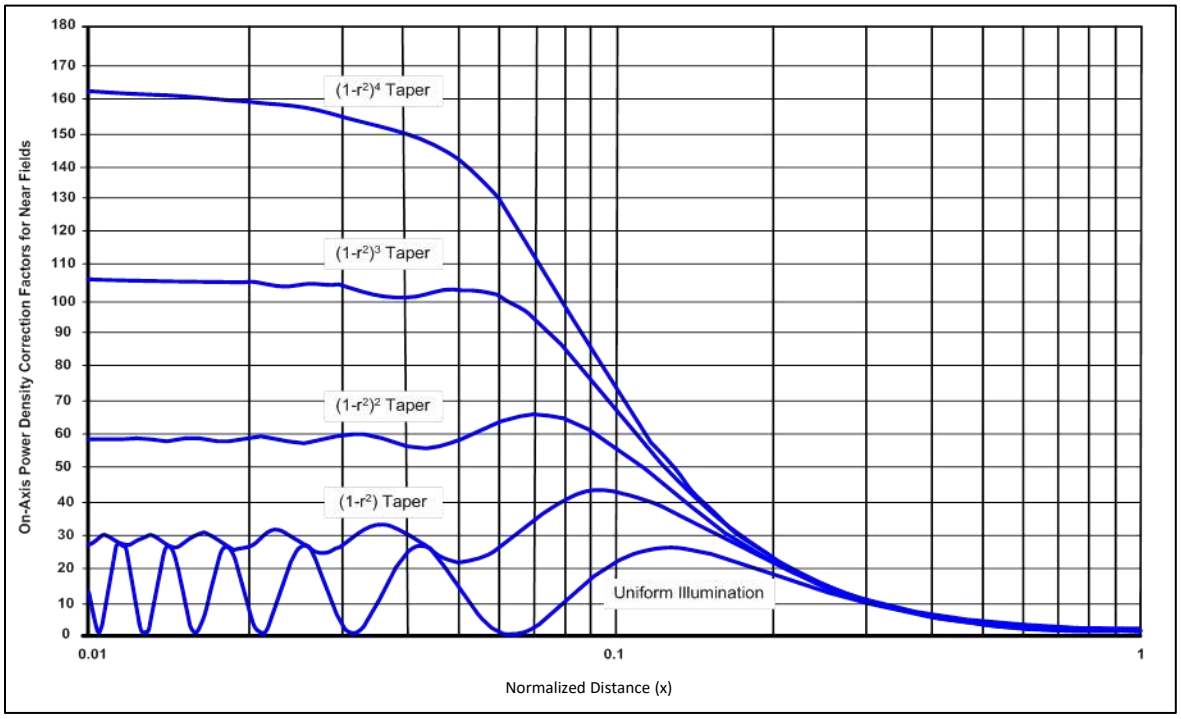


FIGURE 1. Near-field correction factors for circular-aperture illumination

4.1.5.2.2 Near-Field Power Density Calculation Method for Rectangular-Aperture Antennas. Equation (4-14) is used to calculate the power density along the propagation axis of a large-aperture rectangular antenna source in the "radiating" near-field region. The antenna gain and beam width both are degraded in the Fresnel region; therefore, the far-field power density equation (4-7) is modified to account for the near-field antenna gain reduction factor (NGF_{rect}).

$$P_D = \frac{P_t \times G_t}{4 \times \pi \times d^2} \times NGF_{rect} \qquad (4\text{-}14)$$

where:

P_D is the average or peak power density (W/m^2)
P_t is the average or peak transmitter output power (W)
G_t is the numerical antenna gain (unitless)

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d is the distance from the antenna (m)
NGF_{rect} is the near-field gain reduction factor (unitless)

Steps for computing the near-field power density:

a. Compute the power density at the specific distance of concern using the Friis transmission formula (4-7).

b. Calculate the illumination constant (R) for both the horizontal and vertical plane using equation (4-11).

c. Estimate the antenna illumination and antenna factor using the illumination constant and TABLE II.

d. Check to see if the antenna illumination is reasonable by computing the antenna efficiency (K) using equation (4-12).

e. Determine the NGF_{rect} using the distance from the antenna (in wavelengths) and antenna length (in terms of wavelength) using the appropriate antenna illumination graph from FIGURE 2 through FIGURE 6.

f. Multiply the power density (at the distance of concern) computed in step (a) by the NGF_{rect} determined in step (e).

The following provides details for calculating the near-field gain reduction factors (NGF_{rect}) for the Fresnel region of a rectangular-aperture antenna. As stated above, the near-field gain reduction factor depends on the type of antenna illumination and the distance from the antenna. A gain reduction factor for both the horizontal as well as the vertical axis/plane must be determined in order to compute the power density for a rectangular antenna. If the antenna illumination is not known for each axis/plane, it can be estimated by the method described below.

First, calculate the illumination constant (R-value) using equation (4-11) and determine the illumination type and factor (F_h and F_v) using equation (4-15) and TABLE II for each axis/plane. Illuminations above $\cos^4$ are omitted purposely since the gain reduction in the Fresnel region would be almost negligible.

$$F = F_h \times F_v \qquad (4\text{-}15)$$

where:

F is the illumination factor
F_h is the horizontal illumination factor
F_v is the vertical illumination factor

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TABLE II. Rectangular-Aperture Antenna Illuminations

Limits of R	Estimated Illumination	Antenna Illumination Factor (F_h or F_v)
0.88 to 1.2	Uniform	1.00
1.2 to 1.45	cos	0.810
1.45 to 1.66	$\cos^2$	0.667
1.66 to 1.93	$\cos^3$	0.575
1.93 to 2.03	$\cos^4$	0.515

When the illumination constant (R) is determined to be borderline between two orders of illumination, the higher order is selected because it will produce the maximum power density. However, the antenna efficiency should be checked using equation (4-12) to determine if it is reasonable for this antenna type and illumination.

Should the higher-order illumination cause the antenna efficiency to be too high, then the next lower order should be selected. Antenna efficiencies (K) between 0.3 and 0.9 are considered reasonable. Once the illumination type (uniform, cos, $\cos^2$, $\cos^3$, $\cos^4$) has been determined, the near-field gain reduction factor can be selected by normalizing the distance from the antenna and the antenna dimension in terms of wavelength (λ). (See equations (4-16) and (4-17).)

$$\lambda = \frac{300}{f} \tag{4-16}$$

where:

f is the frequency in MHz

$$x_{rect} = \frac{d}{\lambda} \tag{4-17}$$

where:

d is the distance from antenna in meters
λ is the wavelength in meters

FIGURE 2 through FIGURE 6 provide graphical curves of the gain reduction in decibels (dB) versus distance (x_{rect}) from the antenna (in terms of wavelength) relative to the antenna dimension (in terms of wavelength) for each of the illumination types (uniform, cos, $\cos^2$, $\cos^3$, $\cos^4$). Define the aperture dimensions (a_h and a_v) normalized to wavelength as shown in equation (4-18):

$$a_h = \frac{D_h}{\lambda}, a_v = \frac{D_v}{\lambda} \tag{4-18}$$

where:

D_h is the largest horizontal dimension of rectangular antenna in meters
D_v is the largest vertical dimension of rectangular antenna in meters

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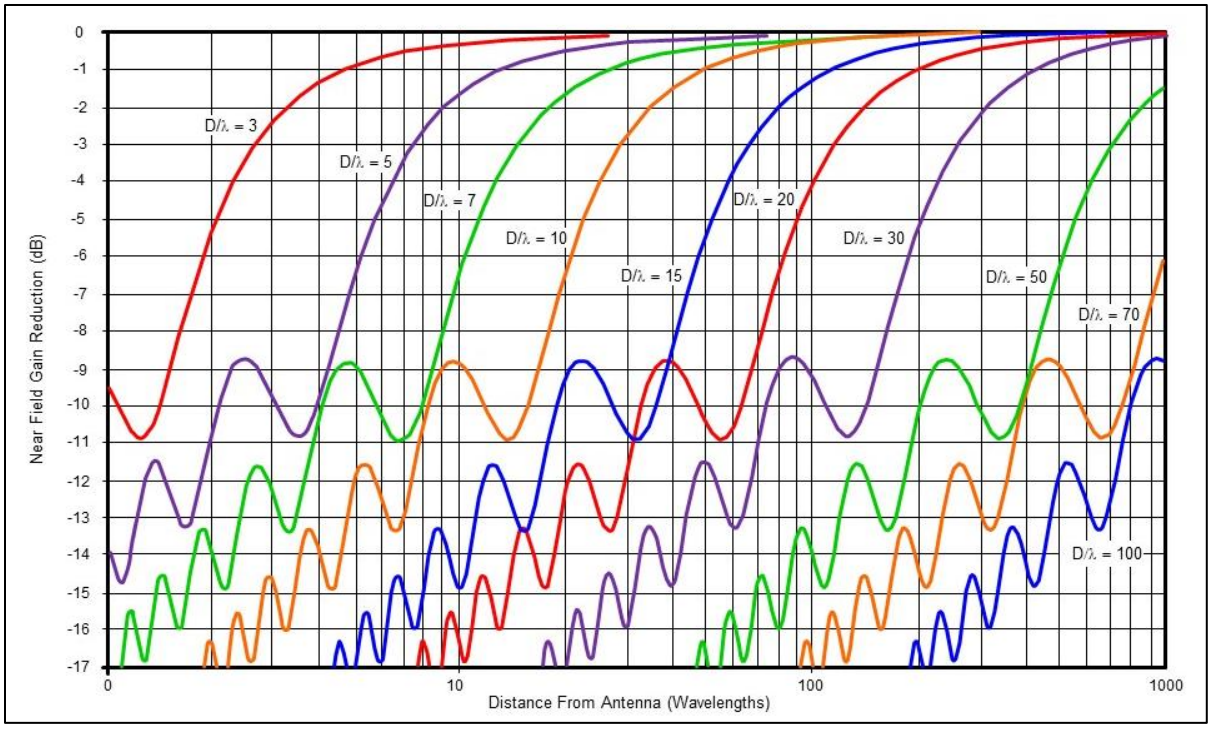


FIGURE 2. Fresnel region gain reduction for uniform illumination (rectangular aperture)

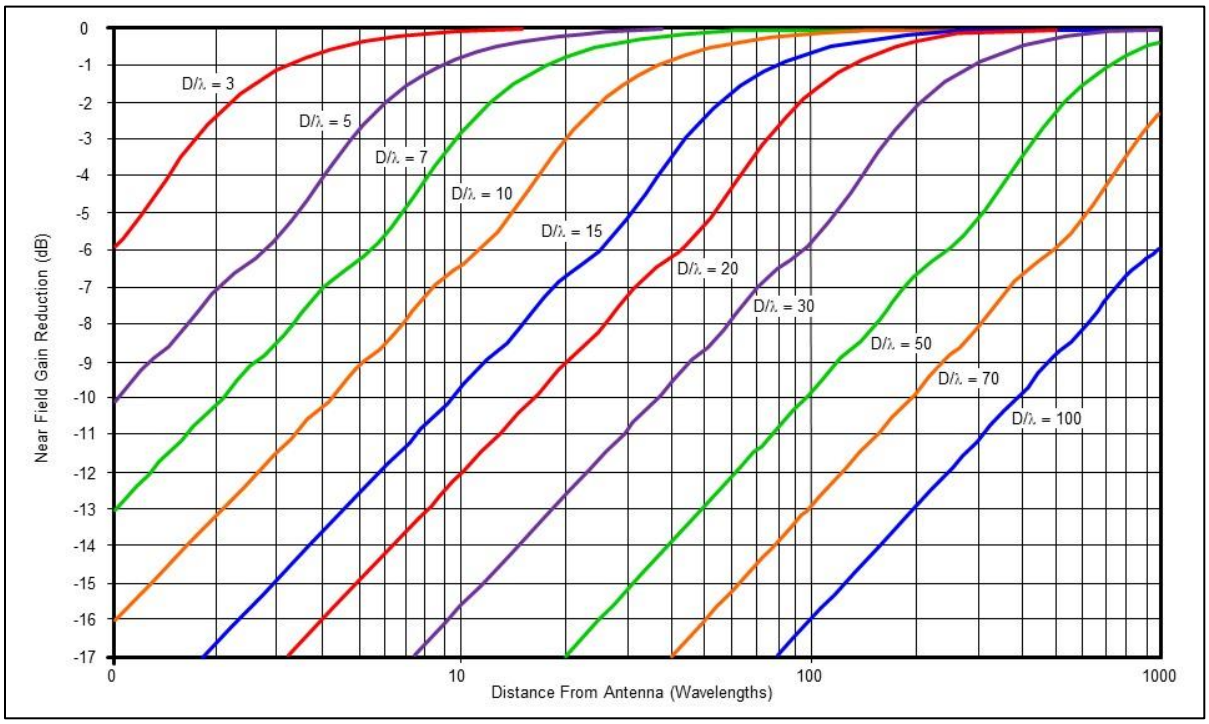


FIGURE 3. Fresnel region gain reduction for cos illumination (rectangular aperture)

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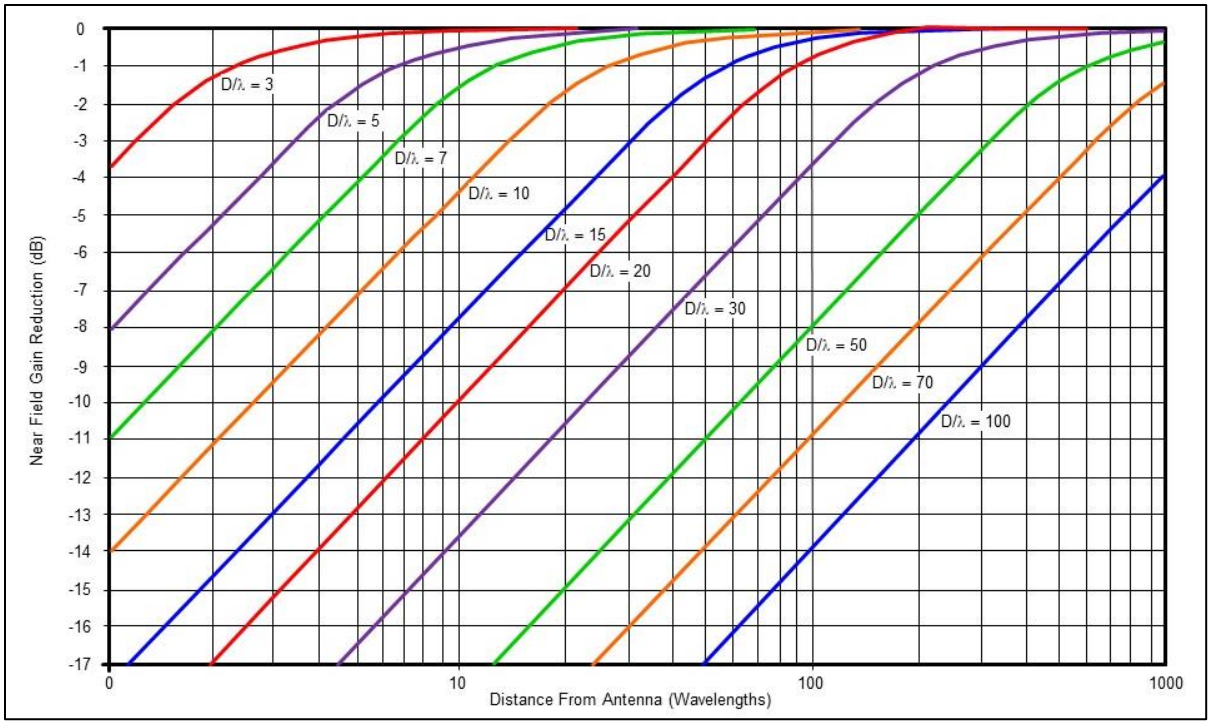


FIGURE 4. Fresnel region gain reduction for $\cos^2$ illumination (rectangular aperture)

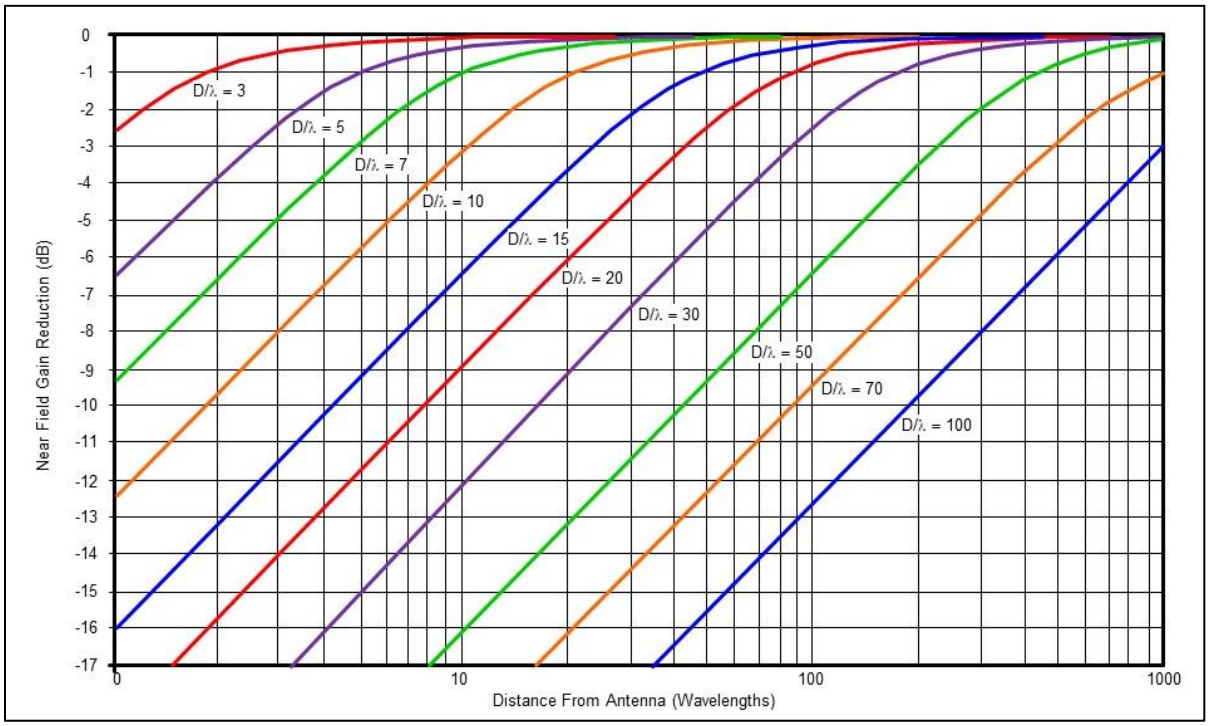


FIGURE 5. Fresnel region gain reduction for $\cos^3$ illumination (rectangular aperture)

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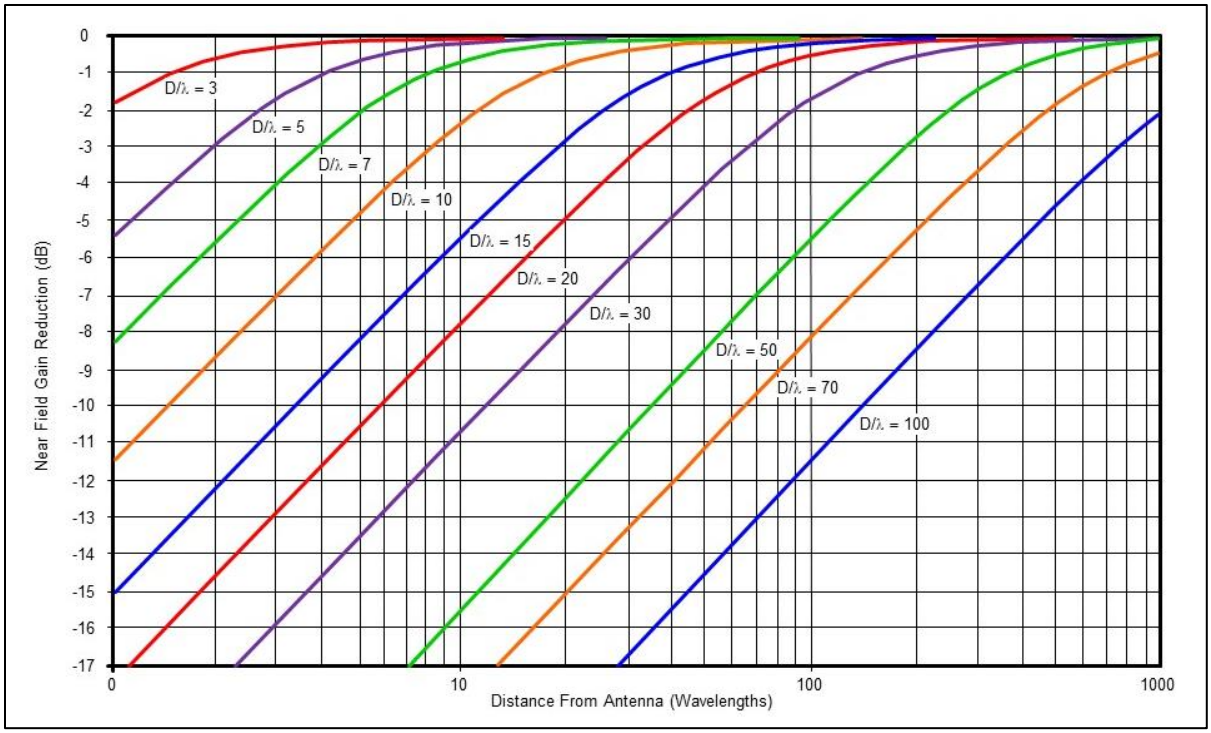


FIGURE 6. Fresnel region gain reduction for cos^4 illumination (rectangular aperture)

Once the two appropriate gain reduction values have been selected for each plane, the combined near-field gain reduction factor (NGF_{rect}) can be computed. To do this, add the two gain reduction values and convert the units from dB to a numerical value. The near-field power density level at a given distance (d) from the antenna can be determined by computing the power density and multiplying this value by the (NGF_{rect}) using equation (4-19).

$$NGF_{rect} = 10^{\frac{NGR_v + NGR_h}{10}} \qquad (4\text{-}19)$$

where:

NGR_v is the near-field gain reduction for vertical axes in dB
NGR_h is the near-field gain reduction for horizontal axes in dB

4.2 HERO Survey Measurements

4.2.1 Introduction. A comprehensive RF radiation hazard assessment or "HERO Assessment" requires a systematic approach and may be accomplished through on-site measurements and/or near-field or far-field calculations of the EME, previously discussed in 4.1. Either technique includes the gathering of pertinent data such as transmitter/antenna configurations and locations, ordnance and platform data, and ordnance locations during the entire stockpile-to-safe-separation sequence (S4). The results and conclusions of the assessment must be clear and concise as they are used to generate HERO operational guidance; for more information see MIL-HDBK-240-7.

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On-site measurements of RF energy to characterize the fields produced by fixed, portable or mobile emitter systems, and interpretation of the resulting data requires a team of trained survey personnel who are knowledgeable of the test equipment and experienced with the various types of emitters. Assessments of the RF environment may also include measurements in enclosed reflective spaces, which have an additive effect on RF, such as when a wireless fidelity (Wi-Fi) system or mobile/portable radio is operated in a transportation trailer, ordnance magazine, or platform.

HERO surveys are conducted in three parts:

a. Pre-survey analysis/coordination.

b. On-site visit.

c. Generation of the results and conclusions, which may be presented in a final report.

An assessment report can include the HERO classification for all ordnance; SSDs for all emitters; a detailed map of the installation, area, or platform illustrating all ordnance and transmitter locations and HERO zones; and techniques and/or instructions designed to eliminate or mitigate HERO concerns.

4.2.2 Pre-Survey Analysis/Coordination.

The level of pre-planning by the HERO survey team will determine how successful the on-site visit will be. The goal of the pre-survey analysis/coordination is to establish the appropriate points of contact, determine areas of interest, gather a list of ordnance that can be or are used, and compile a list of emitter systems to be measured. Pre-planning helps the team generate an order of priorities and a density of measurement points that capture the maximum EMEs that may exceed ordnance MAEs at each ordnance location. Additionally, for large areas such as the flight deck/flight line and hangars, a symmetrical grid of measurement points is used to map the EME.

Information gathered should include the following:

a. A prior HERO survey report (if it exists).

b. Up-to-date facility maps or platform drawings that show ordnance storage areas, ordnance assembly/handling areas, arming/de-arming areas, ordnance transportation routes, and specific transmitters and accurate transmitting antenna locations.

c. For ordnance and ordnance activities, the personnel responsible for overseeing the ordnance and ordnance operations. These individuals may include the explosives safety officer, weapons officer, or the ordnance officer.

d. For emitter information, the personnel responsible for the installation and operation of the emitter systems. These individuals may include the communications systems officer, spectrum manager, frequency manager, electronics materiel officer, or ground electronics maintenance officer.

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e. A list of ordnance that can or will be used at the locations being surveyed. This list should include the Department of Defense Identification Code (DODIC), National Stock Number (NSN), and nomenclature of all ordnance.

f. A list of all transmitting equipment (fixed, transportable, and mobile). This list should include the operating parameters (for example, average power, frequency range, and antenna gain) of each transmitter and antenna.

g. Emitter and ordnance information for any aircraft (fixed and rotary wing) normally operating at the location.

h. Identify and provide the appropriate information changes to ordnance and emitters that is expected to be installed or removed.

i. Tenant/Host/neighbor transmitter and antenna information.

j. HERO emission control procedure information currently in effect.

k. Issues that will affect the measurement of the EME, such as scaffolding around antennas, non-operational antenna/transmitter systems, maintenance actions, the presence and movement of cranes, and radiation restrictions must be considered and documented.

l. Any specific concerns of the location being surveyed so that they can adequately be addressed.

4.2.3 On-Site Measurements. On-site measurements may include measurements to characterize the fields related to fixed emitter systems, platform emitters associated with vehicles or aircraft, and portable and mobile systems. Fixed emitter system measurements are generally performed along a grid layout or using a spatial sweep method. Measurements performed using a grid layout technique are useful for locating areas within a location with relatively undefined boundaries where EMEs exceed predefined levels. The spatial sweep method assesses the entirety of a confined boundary to determine the maximum EME within that boundary. Generally, measurements are not performed for portable and mobile systems because the resultant fields are subject to drastic changes as the transmitters move over and around different soils, terrains, and structures of varying conductivity.

Assessments of the RF environment may also include measurements in enclosed reflective spaces, such as a transportation trailer or ship's ordnance magazine. These spaces should be characterized to ascertain the complex EMEs due to the proliferation of wireless systems, such as Wi-Fi, Bluetooth, and automatic identification technology (AIT). RF transmissions inside a metallic structure reflect off the walls and, in certain areas, may create a high EME because of the additive effect of the RF energy.

These characterizations employ a walk-about technique and are conducted by transmitting a swept CW signal over the frequency band of interest at a nominal amplitude of 0 decibels with reference to 1 milliwatt (mW) (dBm) while simultaneously measuring the received power after reflecting it throughout the space. This is accomplished while walking through the space and

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manipulating both the transmit and receive antennas into as many orientations and aspect angles as possible with the measurement receiver set for Max Hold. This walk-about process creates effective RF energy stirring in the space by a combination of source stirring via the moving transmit antenna and volume sampling via the moving receive antenna.

The goal of the on-site visit is to gather pertinent information to assess the HERO posture of the installation/ship in order to assess risks and provide controls to mitigate HERO including the following:

a. Identifying all ordnance items and associated platforms.

b. Identifying types of ordnance operations and locations (including transportation routes, build-up areas, arm/de-arm areas, and so forth).

c. Identifying all antenna/transmitter systems (stationary, mobile, portable, aircraft, and so forth).

d. Identifying antenna/transmitter characteristics (gain, power, frequency, and so forth).

e. Identifying antenna/transmitter locations.

f. Identifying mission requirements

g. Reviewing existing HERO controls, if available.

The on-site measurements start with a visual examination of all ordnance location(s) and the ordnance S4 phase at each location is determined. All transmitting systems (specifications and locations) are identified and the expected EMEs produced by these systems are calculated and compared to the ordnance MAEs with respect to the ordnance S4 phase(s). During this appraisal, it may be advantageous to identify areas (or zones) such that emission control procedures can be minimized to reduce operational restrictions. Finally, the operational EMEs from radar, EW, communications, and navigation transmitters are measured at the locations where those EMEs may exceed ordnance MAEs.

It must be noted that there are several types of measurement techniques that can be used to characterize the EME (for example, flight deck, magazine areas, and so forth) to include the following:

a. Grid mapping of the ordnance areas.

b. Sampling Measurements: 2-3 measurement locations are selected within a large location [or search for a maximum EME to apply as the maximum for that area (for example, parking areas or shipboard topside areas)] to ensure no ordnance MAE is exceeded.

c. Axial Measurements: Measurements are taken along a straight axis from a system's antenna towards an ordnance location until the measured EME is less than either the general or ordnance specific MAEs.

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d. Validation Measurements: Used to validate a calculated SSD or demonstrate levels are below a defined MAE from an antenna.

Each of these measurement techniques has its merits and should be considered prior to the survey to determine the best technique to support control measures that offer the least restrictions to operations.

Prior to conducting measurements, there are a number of things to consider:

a. Frequency authorization should be coordinated to ensure that the assessment team can freely exercise the transmitters. This should be done prior to the on-site visit and coordinated with the local frequency manager.

b. All modes of operation, waveforms and the expected EMEs for all systems being measured should be fully understood to ensure the measurement is valid and the worst-case values are measured.

c. Access to transmitters and local operators should be made available prior to conducting measurements.

d. Identify impediments to characterizing the EME including, but not limited to, scaffolding around antennas, non-operational antenna/transmitter systems, maintenance of ordnance operation areas, nearby ordnance operations, obstructions, reflections, maintenance and construction activities, radiation restrictions, nesting of platforms, and proximity of cranes. All of these can either impede the measurement process or affect the EME to be measured.

4.2.4 Equipment Requirements. A representative suite of assessment equipment for HERO measurements are shown in TABLE III. The assessment team should ensure that the equipment covers the dynamic range of the spectrum so that the full EME is captured/recorded during the assessment. It may be beneficial that the equipment used for assessment measurements have the same bandwidth and specifications as the equipment used for establishing field measurements during HERO testing of ordnance. This dual use provides the HERO programs a direct correlation of operational EME to the ordnance response. Equipment should be within current calibration with the standards traceable to the National Institute of Standards and Technology (NIST).

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TABLE III. Measurement Equipment

Equipment	Description/Type
Power meter	A device used to measure power received or delivered to a transmitter/receiver.
Spectrum analyzer(s)	An RF measurement device that displays power and spectral content of input signals typically over large bandwidths. Requires ancillary equipment such as antennas and interconnecting cables.
Isotropic radiation monitor	An RF measurement device that utilizes an omni-directional antenna. The antenna is typically integrated into one handheld package with a measurement display.
E-field probe(s)	A directional or omni-directional antenna used to measure E-fields. Fiber-optic outputs can deliver the measurement data to the receiver for processing, thereby reducing interference caused by coupling of EMEs to cabling.
H-field probe(s)	An antenna, typically constructed from a wire loop, used to measure H-fields. Fiber-optic outputs can deliver the measurement data to the receiver for processing, thereby reducing interference caused by coupling of EMEs to cabling.
RF attenuator	Typically installed between an antenna and a measurement device, to suppress input signals to RF measurement devices for overload protection.
Shielded enclosure	An overall encapsulation for an electronic device made of conductive materials used to protect electronics from EMEs. Ideally, the enclosure is continuous and fully encompasses the device and wiring.

4.2.4.1 Equipment Measurement Precautions. One of the problems of measuring high-level electromagnetic fields is false indications due to coupling of the field into the measurement equipment. A termination connected to the input of the instrument can be used to determine if electromagnetic energy penetrates the case and produces a false indication on the meter, or if it affects the calibration setting. It may be necessary to place the equipment in a shielded enclosure or use a long coaxial cable to connect the antenna to the meter placed at a location with sufficiently lower electromagnetic fields.

The selection of RF measuring equipment used to make field intensity measurements depends on the type of emission being measured. Four measuring systems are described:

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a. Power measurement system.

b. Spectrum analyzer.

c. Isotropic (field) probes or power density meters.

d. Isotropic radiation monitor.

Excessive signal input power will damage the spectrum analyzer and/or power meter system. Therefore, precautions must be taken to protect the measuring equipment by installing in-line external attenuators.

4.2.4.1.1 Power Measurement System. Power meter/sensors are used with test antennas to measure power. The key advantage of a power sensor is the instantaneous bandwidth available. It can capture full peak or average power of an emitter in real time. Power sensors come in many frequency ranges, amplitude ranges and detector types. Some sensors require an associated display and newer ones can connect to computers to view and analyze measurement results.

Some limitations are:

a. No frequency selectivity: The highest signal strength at the test location is displayed irrespective of frequency, possibly resulting in data recorded for the wrong system.

b. The sensitivity is reduced by the large bandwidth, making detection of low signal strengths difficult or impossible.

c. A maximum hold must be used when measuring moving or phased-array antennas.

d. The response time could be longer than the pulse width of the emitter or beam switching of a phased array. This could cause inaccurate measurements.

4.2.4.1.2 Spectrum Analyzer (Signal Analyzer). The spectrum analyzer has a key advantage of being frequency selective. It can be used in complex environments to focus on specific emitters and to identify unknown or poorly tuned emitters. Traditional swept tuned analyzers may require many cumulative sweeps to capture the peak power of pulsed emitters and/or antennas in motion. Newer Fast Fourier Transform (FFT)-type analyzers can significantly reduce the capture time required.

The resolution bandwidth must be large enough to avoid desensitization of the measurement or a correction must be made.

Most analyzers have multiple detector and trace options. Average detector and average trace modes may not yield true average power measurements. The peak power must be converted by using the duty cycle of the transmitter to calculate average power.

In order to meet specified frequency and amplitude accuracy, the analyzer's manual calibration procedures must be performed each time the analyzer is powered up and periodically as needed.

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Warm-up times may also be required. Refer to the analyzer operation manual for specific procedures.

External attenuation must be used whenever the total RF power into the spectrum analyzer exceeds the specified maximum input level (a common limit is +30 dBm).

4.2.4.1.3 Isotropic (Field) Probes or Power Density Meters. These are similar to power meters in function. They typically perform average power measurements.

Some equipment limitations are as follows:

a. No frequency selectivity: Ensure that the system being tested is the only system transmitting or is producing a high field strength when compared to other transmissions at the test location.

b. The sensitivity is limited by the large bandwidth, making lower level detections more difficult.

c. A maximum hold must be used when measuring moving or phased-array antennas.

d. The time response of a probe could be longer than the pulse width of the emitter or beam switching of a phased array, which has the potential to cause inaccurate measurements.

e. Use next to metallic structures has the potential to produce inaccurate measurements.

4.2.4.1.4 Isotropic Radiation Monitors. The isotropic radiation monitors are employed in complementary roles with the power meter/spectrum analyzer. They allow a quick look at the general EME.

4.2.4.2 Antenna Measurement Procedures. A communications link between the transmitter operator and survey team will need to be established. After all measurement equipment is in place, emitters are then energized, and the survey team will take measurements and record all necessary data at each of the assessment locations for each antenna. It may be necessary to selectively silence other emitters while a particular antenna is being measured.

The output power of the transmitter may be measured with a power meter through a directional coupler between the transmitter and the antenna.

The following formula is used to calculate the power density from the measured data:

$$P_D = P_R + L + a - A_e \qquad (4\text{-}20)$$

where:

P_D is the power density (dBm/cm²)
P_R is the power received (dBm)
a is the resolution band width correction factor (dB)

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L is the cable losses (dB expressed as a negative value)
A_e is the effective area of the receive antenna (dB/cm^2)

Convert to mW/cm^2:

$$P_D(mW/cm^2) = 10^{\left(\frac{P_D}{10}\right)} \quad (4\text{-}21)$$

Convert the peak mW/cm^2 reading to an average mW/cm^2 reading, using the formula:

$$Average\ P_D(mW/cm^2) = Peak\ P_D(mW/cm^2) \times duty\ cycle \quad (4\text{-}22)$$

4.2.4.2.1 Mechanically Rotated Antenna Power Density Measurements. In order to meet the requirements of radiation restrictions under certain circumstances, antenna rotation may need to be stopped in a non-interfering direction with the transmitter operating in its normal mode. Power density measurements will be taken along the line of radiation.

It may not always be possible to silence all emitters in the frequency range of the receive antenna nor stop antenna rotation of systems undergoing measurement. Therefore, to capture the full environment the user will have to use the correct equipment to capture and dwell longer during the measurements.

In both cases, the receiving antenna will be utilized so as to obtain a representative sample. "Worst-case" will be recorded as a datum at each particular assessment point.

4.2.4.2.2 Electrically Steered Antenna Power Density Measurements. Take power density measurements in and around all affected ordnance areas. The receiving antenna will be utilized so as to obtain a representative sample. "Worst-case" will be recorded as a datum at each particular assessment point.

4.2.4.2.3 Directed-Beam Power Density Measurements. Take measurements where the director/illuminator or other radars can directly illuminate ordnance areas.

When measuring unknown EME levels, the procedure below will be followed:

With personnel in a safe area and the receiver antenna on a tripod positioned in the location at which measurement is to be made, visually align the antenna being measured with the receiver antenna and radiate. While radiating, move the transmitting antenna in azimuth and elevation (as applicable to the system being measured) until a peak measurement is obtained, and record that antenna location with the measurement data. Increase the antenna elevation angle and record the measurement data until it is below hazardous levels.

4.2.4.3 Data Reduction and Evaluation. A detailed data reduction is performed to identify sources of potential hazardous EMEs that exceed HERO MAEs at ordnance operations areas and the locations where excessive EMEs occur. It will also be reviewed for any unexpected results.

TABLE IV lists all data reduction variables.

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TABLE IV. Data Reduction Variables

Equipment	Variable
Power meter	Meter reading Coaxial cable loss In-line external attenuation Transmitter rated/actual power
Spectrum analyzer	Amplitude reading (dBm) Center frequency Frequency of peak amplitude reading Band width Resolution band width Coaxial cable loss In-line external attenuation Transmitter rated/actual power Transmitter duty cycle

4.2.5 Complex Cavity Measurements. EME measurements are also conducted inside reverberant spaces to characterize the complex EMEs that potentially could be produced inside these cavities by the wireless systems currently being used. The goal of these measurements is to determine the space's cavity calibration factor (CCF), which is used to estimate the peak E-fields from the total power radiated in the space. Guidelines for allowable total transmitter powers for radio frequency identification (RFID), wireless local area network (WLAN), and Wi-Fi systems, as well as any other transmitters that can be used within below-deck spaces, then can be established based upon the results of these measurements. There are multiple methods that can be applied for measuring the CCF of a cavity. Traditional methods are based on calibrations of reverberation chambers and involve utilizing mechanical tuners to change boundary conditions and adequately stir the RF energy within the space. However, an alternative method was developed, referred to as the walk-about technique, which leverages source stirring and spatial sampling. This method can be conducted without the need for constructing mechanical tuners within the cavity of interest, thereby significantly decreasing the time required to perform the measurement.

4.2.5.1 Walk-About Technique. The "walk about" technique is implemented using a hand-held spectrum analyzer operated in the tracking generator mode (or alternatively a vector network analyzer) and connected to a pair of antennas (transmit and receive) mounted cross-polarized on a pole. TABLE V lists the general spectrum analyzer settings for use during the measurement. CCF measurements are conducted by transmitting a swept CW signal over the frequency band of interest at a nominal amplitude of 0 dBm (1 mW) while simultaneously measuring the received power after the RF has reflected throughout the space. This is accomplished by walking through the space with the measurement receiver set for Max Hold. The random walk should be performed for five minutes and upon completion, the resultant trace is stored. The process is then repeated to obtain a total of 12 traces. A calibration measurement should also be performed with the transmit and receive antennas removed and cables connected to one another to characterize the system losses. Following measurements, the calibration data will be used to correct for the cable attenuation, thereby providing a true measure of the volumetric insertion loss for each space.

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TABLE V. Spectrum Analyzer General Settings

Spectrum Analyzer/Tracking	Value
Frequency span	400-6000 MHz
Reference level	0 dBm
RF attenuator (receive)	0 dB
Tracking Generator on (transmit	0 dBm
Pre-amp	Off
Resolution bandwidth	100 kHz
Video bandwidth	3 MHz
Sweep time	2 sec
Sweep mode	Free run (continuous)
Trace mode	Max hold
Detection	Sample

4.2.5.2 Data Processing. FIGURE 7 shows typical corrected maximum received power data that have been averaged over the 12 runs, and then compared to maximum and minimum values. The average is a frequency-dependent value that represents space insertion loss. FIGURE 8 depicts the uniformity of the data, which provides a quick and easily understood way to determine measurement quality and insight into E-field amplitude variability due to RF sources placed within enclosed spaces. This plot specifically shows the extent of maximum and minimum power normalized to the mean of the max received power. A large spread (greater than ±3 dB) indicates 1) cavity size is insufficient to support uniform propagation of that frequency, or 2) a data collection anomaly. Since greater variations from the mean indicate a decreased probability that the measured maximum will be observed in that space, the risk of electromagnetic interference (EMI) due to reverberant energy can be quantified accordingly.

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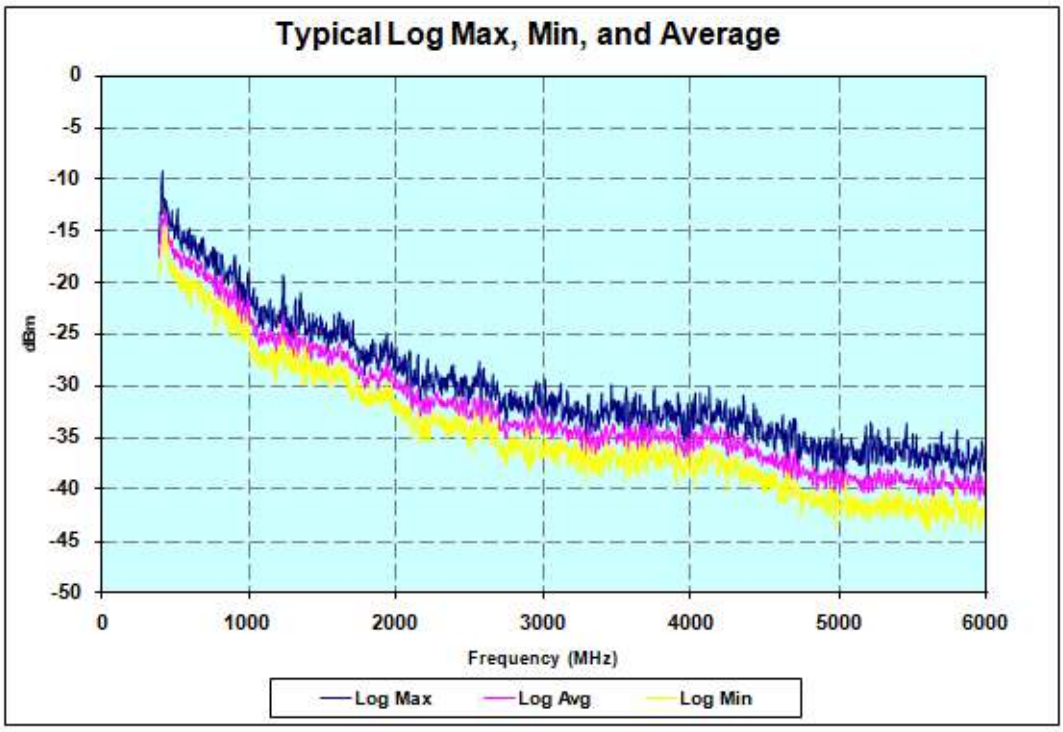


FIGURE 7. Typical Max, Min, and Average Insertion Loss

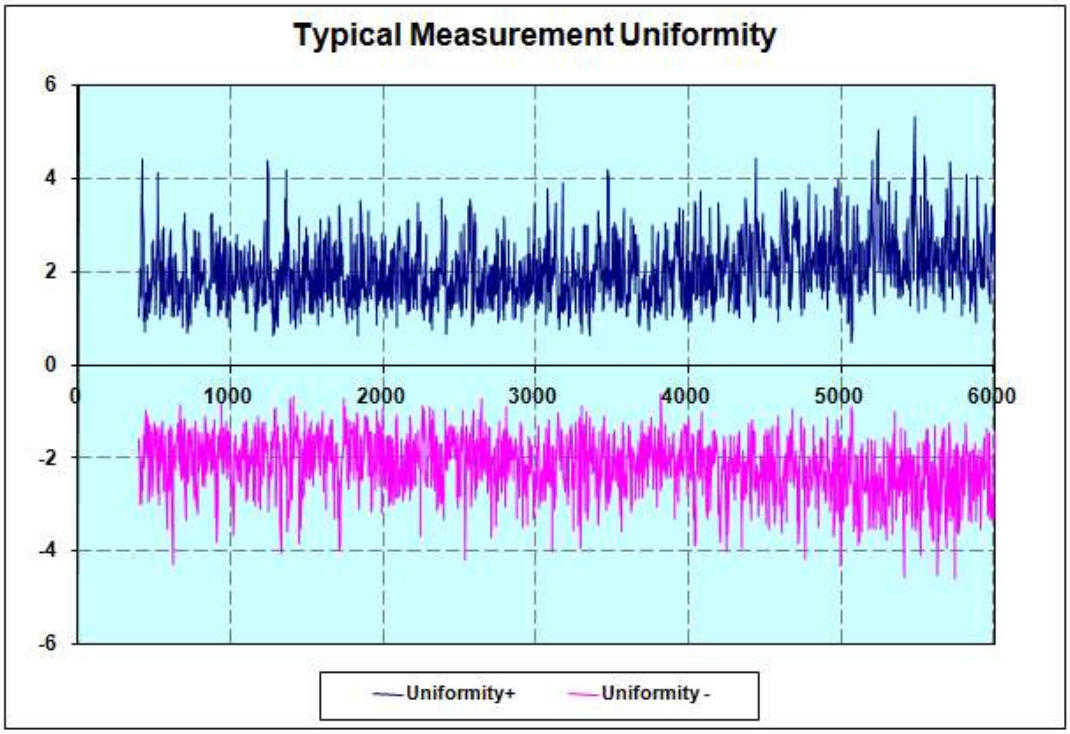


FIGURE 8. Typical Uniformity

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A *CCF* or normalized E-field in V/m/√(W), shown in FIGURE 9, provides the relationship of the power input into the space and the corresponding maximum expected EME within the space via equation (4-23);

$$CCF \equiv Normalized\ E_{Max} = \frac{E_{Max}}{\sqrt{P_{input}}} - \frac{8\pi}{\lambda}\sqrt{\frac{5\times IL}{\eta_{rx}}} \qquad (4\text{-}23)$$

where the insertion loss (IL) is the quotient of the measured maximum power to the given total input power emitted into the space,

$$IL = \frac{P_{max\ (rec\ per\ run)}}{P_{input}} \qquad (4\text{-}24)$$

and η_{rx} is the efficiency of the receive antenna. Once the CCF has been determined for a space, the maximum E-field component amplitude in that space can be predicted. For source-victim separations greater than a critical distance governed by free-space loss and antenna directivity, the maximum E-field at the victim will be dominated by the diffuse (reverberant) field, and can be estimated by,

$$E_{Max} \cong CCF\sqrt{P_{input}} \qquad (4\text{-}25)$$

where P_{input} is the total amount of power radiated into the space at the given frequency. (P_{input} is the summation of radiated power from each source in case of multiple emitters at a given frequency.)

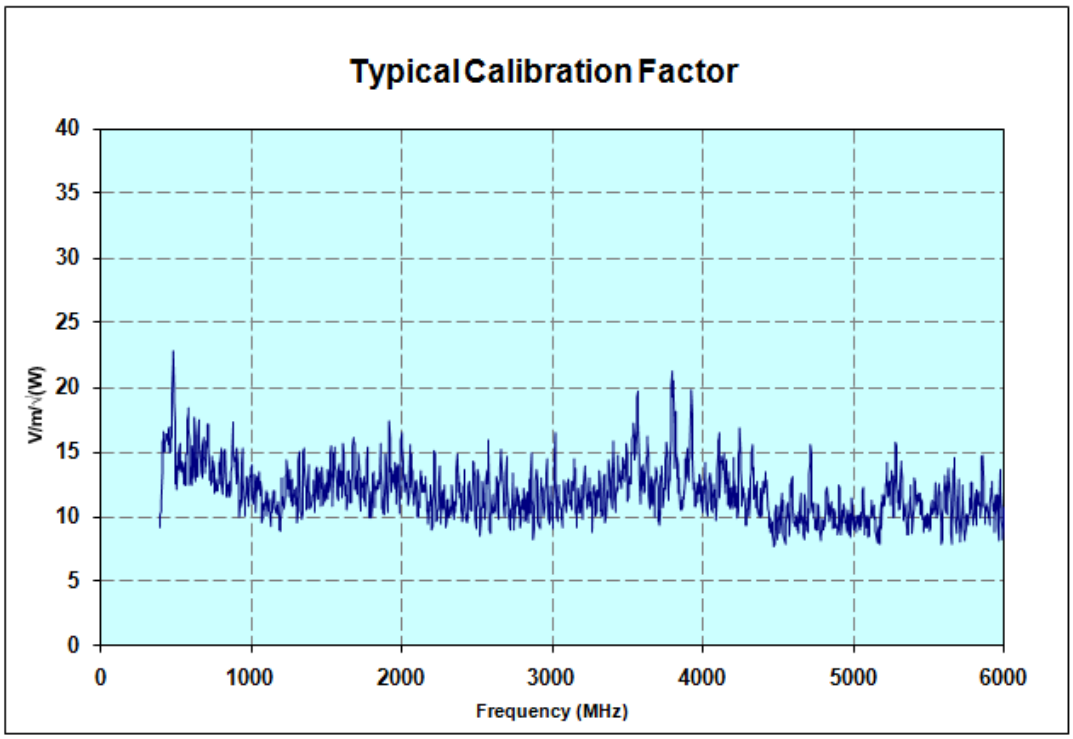


FIGURE 9. Typical CCF

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4.2.5.3 Analysis of EME. A maximum-envelope fit to the CCF data provides a conservative estimate of the CCF across the full frequency range. For example, in one particular space with radiated emission at 2.4 GHz, CCF ≈ 10 V/m/√(W). Hence, the maximum E-field environment at 2.4 GHz is estimated from (4-25) and shown in FIGURE 10.

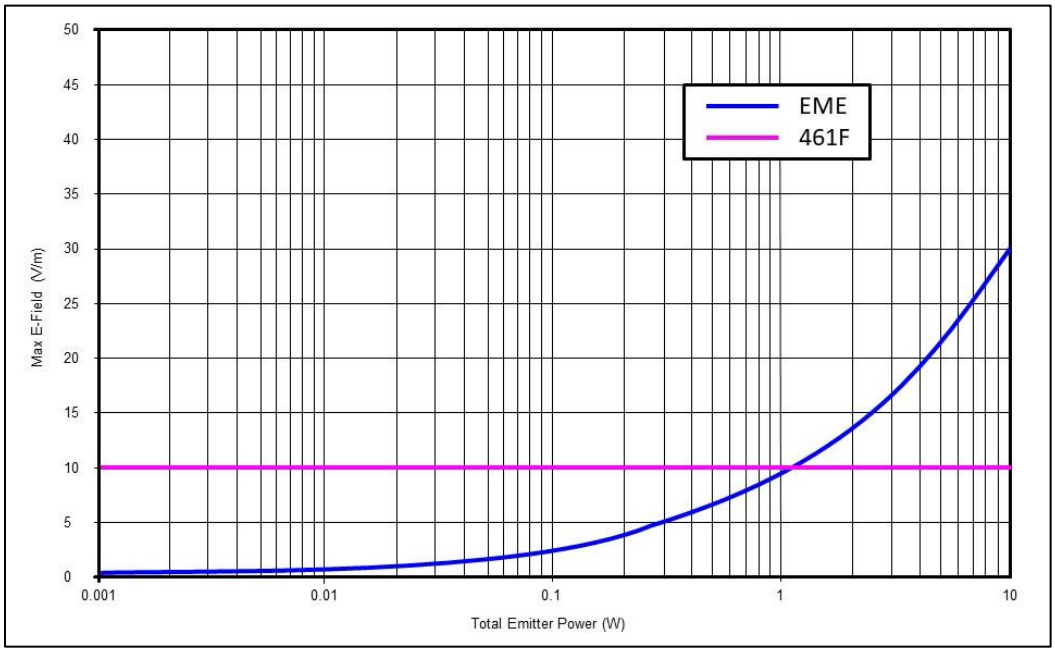


FIGURE 10. EME in a space at 2.4 GHz

Finally, if multiple sources are emitting at different frequencies, the superimposed maximum field environment can be approximated as the root-sum-squares (RSS) of the maximum fields established at each frequency by the sources. The MAE for ordnance is specified and sets an upper limit on the total maximum power emitted into the space.

4.2.6 Ground Mobile Surveys

4.2.6.1 Introduction. As part of the E3 evaluation program for ground mobile systems, platform surveys can be conducted to identify and mitigate HERO and other radiation hazard (RADHAZ) issues.

4.2.6.2 Background. Due to the proximity of the vehicular mounted antennas to ordnance and personnel, it is not always possible to maintain SSDs and a more accurate and precise means for mitigating HERO is required. The test approach describes the HERO measurements conducted for the vehicle, requirements, assessment methods, and procedures required to complete the assessment efforts. This involves defining maximum EMEs based on relevant ordnance MAEs to plan data gathering efforts directly in and around the vehicle.

4.2.6.3 Test Point Selection. The HERO measurements require that field strengths be measured around, on top, and inside the vehicle. The measurement data should be compared to the MAEs for HERO SUSCEPTIBLE ORDNANCE and HERO UNSAFE ORDNANCE to determine the potential for HERO.

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FIGURE 11 depicts sample vehicle antenna locations and measurement test points around the perimeter of the vehicle. Test points 1 through 10 are 12 feet (3.5 meters) from the vehicle frame. This distance was determined based on the modeling of a worst-case ordnance staging or handling area. Measurements should be taken over a range of heights as the measurement antenna is scanned in the vertical direction.

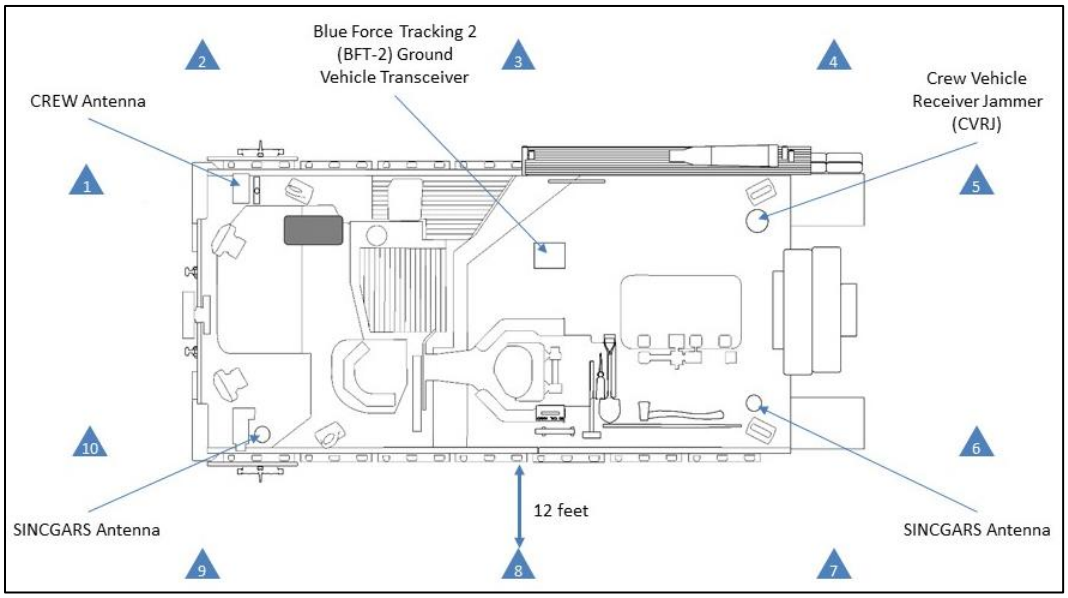


FIGURE 11. HERO test points in relation to a vehicle

In addition to test points 1 through 10, high-frequency (HF) transmitter system testing should include measurement test points along four radial directions around the vehicle, as shown in FIGURE 12. Measurements should be taken in 25-foot (7.5-meter) increments and at the precise locations where the HERO UNSAFE ORDNANCE MAE (0.5 V/m) derived from MIL-HDBK-240-7 are exceeded. For HERO measurements, all HF whip antenna configurations should be considered.

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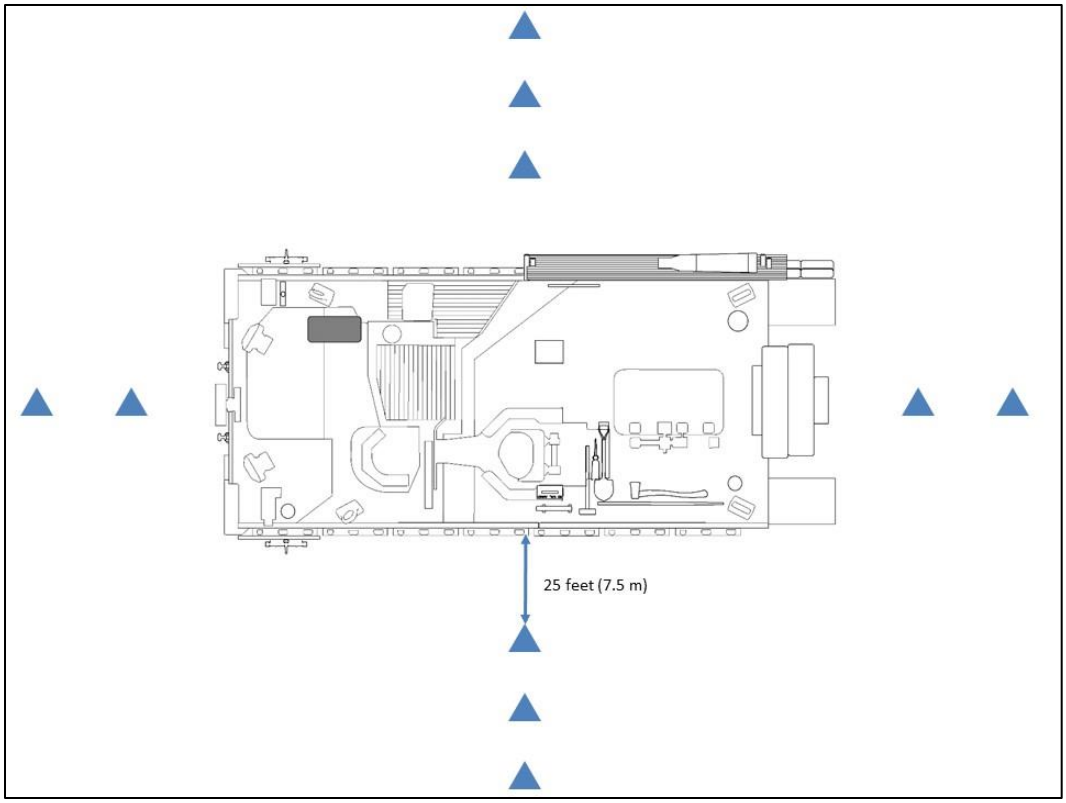


FIGURE 12. HERO measurement test points for HF radials

HF measurement test points should include the topside of the vehicle at ordnance locations. The size of the vehicle and number of apertures are used to determine the number of test points.

A typical vehicle network-centric setup of four emitters would dictate that 40-plus field-strength measurements be taken 12 feet (3.5 meters) from the vehicle under test (VUT) plus topside of the VUT, as depicted in FIGURE 11. These measurements include testing each individual transmitting system while radiating, and then with all transmitting systems radiating simultaneously. The combined fields will account for superimposed effects.

All test points thus far have been used to account for the EME outside the vehicle. Inside the vehicle, measurements are taken at all ordnance stowage and handling areas with all vehicle antennas radiating. It may be necessary to measure each transmitter's resultant EME individually in order to isolate the transmitters that exceed HERO MAEs.

Space limitations within a vehicle often require the need to use smaller power density meters with isotropic E-field probes to measure the resultant EME. The E-field probe is carefully maneuvered within the ordnance space to capture the highest EME. It is recommended that the probe be kept a minimum of 18 inches from conductive surfaces, if possible, with the caveat that measurements within this distance generally result in higher measured values.

4.2.6.4 Prior to Performing Measurements. Identify and record all transmitter systems installed aboard the vehicle. During this phase of testing, appropriate support personnel are

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required to be present to ensure identification of transmitters and antennas, proper operation, and excitation of all onboard RF emitters.

Expected output and VSWR of each communications system should be verified, measured, and recorded.

4.2.6.5 Exterior Measurements. The first phase of testing requires calibrated antenna field-strength measurements at the pre-determined test points. These measurements are intended to give the most accurate field-strength data to determine SSDs. All transmitter systems are measured 12 feet (3.5 meters) from the VUT with either an HF field-strength meter or a spectrum analyzer connected to a receive antenna, as shown in FIGURE 13. A biconical receive antenna with balun (or equivalent) is used for the 30 to 1000 MHz frequency range, while a dual-ridged horn receive antenna is used for frequencies greater than 1000 MHz. The receive antenna should be scanned vertically in multiple polarizations with the spectrum analyzer configured for a maximum hold measurement. For all measurements, the maximum power should be measured and recorded. Special care should be taken to account for a worst-case polarization.



FIGURE 13. HERO measurements 12 feet from vehicle

For the HF system configuration, field-strength measurements should be taken with the whip antenna in the position, that generates the worst-case field strengths. Additional HF measurements should be taken along the four radials as depicted in FIGURE 12.

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FIGURE 14. HF forward radial measurement

4.2.6.6 Interior Measurements. The second phase of testing should consist of taking field-strength measurements in and on the vehicle. First, a test engineer should use an E-field probe using the measurement methodology from IEEE STD 1309-2013 (incorporating a geometric mean). The measurement technique includes 360-degree rotation of the probe in orthogonal orientations. (Note: With the possible exception of transmitter operators, personnel generally are not permitted in occupational areas during measurements.) This technique allows the accurate detection of hot spots and determines the spatial electromagnetic field at any point. One spatial field point must be accomplished for each occupational area. (For example, at a minimum, one spatial measurement can be taken at the driver's seat area; however, multiple peak measurements can be taken at any given point within the occupational area.) Further field-strength measurements are taken for each transmitting antenna. For all measurements, the maximum power should be measured and recorded.

4.2.7 Vehicle Configuration. Any vehicle modifications that may be necessary during testing should have the approval of the appropriate Government representatives on site. No section/component of any platform under test will carry any ordnance with propellants, squibs, or other EIDs. Any new configuration (that is, transmitter addition, antenna relocation, armor retrofits, and so forth) will warrant new measurements and/or analysis. All such modifications must be documented.

4.2.8 Test Equipment/Ancillary Equipment

The following test equipment/ancillary equipment (or equivalent) will be required for the test:

a. Spectrum analyzer

b. HF-very high frequency (VHF) broadband balun

c. Biconical elements

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d. Double-ridged guide antenna

e. HF field-strength meter with whip antenna

f. Wattmeter, 50 Ω impedance

g. Dummy load, 50 Ω impedance

h. E-field meter with probe

i. Touch probe

j. N-type cables

k. 20-dB attenuators (10 each)

l. Rod antenna with tripod and counterpoise

The following ancillary equipment will be used to stimulate the SUT, if necessary:

a. Antenna dipole kit

b. Signal generator

All items must be confirmed with current calibration labels, and all cables and attenuators should be calibrated for attenuation loss.

4.2.9 Test Procedures. The following steps represent sample test procedures. Procedures may be modified at the discretion of the test director/conductor.

a. Provide daily safety brief/plan of the day.

b. Conduct inventory/visual inspection of antennas/systems (nomenclature of transmitters/antennas).

c. Verify output powers and checkouts for RF systems (conduct VSWR and record).

d. Set up calibrated antenna/probe 12 feet (3.5m) away from the vehicle. Map out test points according to 4.2.6.3. Take spectral measurements at test points while scanning the receive antenna in the vertical direction.

e. Determine personnel exposures at occupational areas inside the cabin. Measure any surface currents inside the cabin area, as required.

f. Determine personnel exposures in the cupola/outer hatch areas. Ensure that all antennas have been evaluated in every antenna configuration.

g. Repeat steps d through f for the next antenna system.

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h. During HF system testing, assign four radials (as shown in FIGURE 12).

i. Measure the HF system in 25-foot (7.5m) increments along radials (worst-case frequencies allowed by the test facility).

j. Take additional distance measurements at HERO UNSAFE ORDNANCE and HERO SUSCEPTIBLE ORDNANCE field strengths (0.5 V/m and 2.0 V/m, respectively).

Pretest analysis of emitter characteristics can be accomplished using the MAE Analysis Program. FIGURE 15 depicts a sample graphical display of the HF Transmitter generated using the MAE Analysis Program.

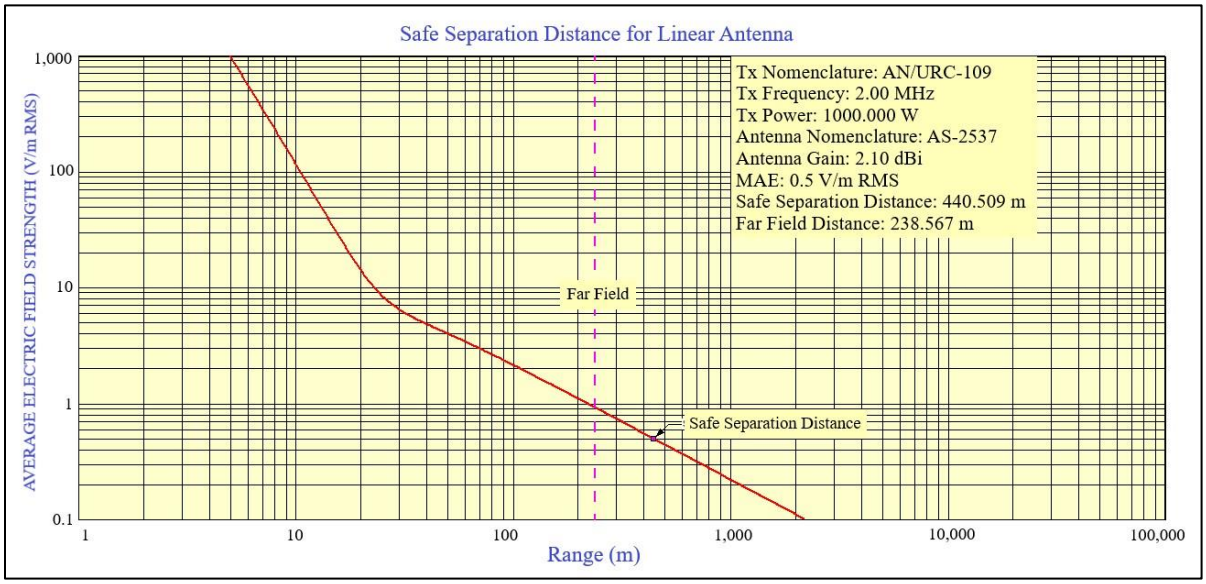


FIGURE 15. Sample graphical output of an HF transmitter simulation

A sample SSD for HERO UNSAFE ORDNANCE is shown as 1445 feet (440 meters) at 1000 W HF, which does not account for reflections, efficiency of the system, and any gain associated with the vehicle.

4.2.10 Aircraft Platforms.

4.2.10.1 Introduction. Aircraft platforms may be integrated with emitters such as VHF and ultra high frequency (UHF) communications systems, as well as, satellite communications (SATCOM) systems, navigation systems, radars, and electronic warfare (jammers). As part of the E3 evaluation program for aircraft systems, EME surveys and intra-system HERO tests should be conducted to identify and mitigate HERO risks. The following paragraphs provide recommendations like those for ground mobile systems discussed in 4.2.6. Knowledge of the field-strength levels at specific ordnance locations is required to determine if the levels exceed applicable MAEs, either for the general HERO UNSAFE or HERO SUSCEPTIBLE levels (see Section 4, Part 7 of this document) or for specific test-derived MAEs. Comparison of the actual radiated field strengths to the MAEs can be used to identify the presence and degree of potential HERO risk. Field strengths should be determined at or near ordnance attachment locations and

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in areas external to the aircraft where ordnance is handled or staged. In either case, field levels may exceed MAEs and thus pose a HERO risk, prompting risk mitigation approaches; for example, silencing the emitter or limiting output power or, for unattached ordnance, increasing the separation distance to reduce incident fields to acceptable levels--that is, reach an SSD.

4.2.10.2 Background. Maintaining SSDs between aircraft-mounted antennas and aircraft-loaded ordnance and other systems containing EIDs (such as chaff/flare dispensers, inflatable vests, and so forth) may pose challenges like those for ground vehicles. Aircraft antenna locations are chosen to achieve desired radiation patterns and ordnance locations are selected to ensure weapon effectiveness and satisfy structural requirements. An assessment of the fields produced by onboard emitters at (or near) aircraft ordnance locations and in areas external to the aircraft where ordnance may be handled and/or staged, is crucial to determine if there are potential HERO risks.

4.2.10.3 Approach. The preferred approach is to conduct an *intra-system HERO test* to quantify the effect of onboard emissions by monitoring instrumented EIDs as the aircraft emitters are operated. An intra-system HERO test is, in many ways, like the HERO testing described in Part 2 of this handbook, with the key difference being that, in intra-system testing, the test environment is produced by the aircraft emitters, vice external transmitters and antennas at a HERO test facility. The focus in intra-system testing is on handling/loading, platform-loaded, and staged S4 configurations and ordnance procedures. Additional guidance applicable to EID instrumentation and ordnance handling/loading procedures may be found in Part 2 of this handbook. The advantage of intra-system testing is that operational field-strength levels and coupling of electromagnetic energy to the ordnance are most accurately represented when the ordnance is subjected to the actual environment produced by transmitting aircraft emitters. Field blockages, aspect angles, polarization, and superposition effects that can influence energy coupled to the EID are produced during intra-system tests.

A less desirable alternative (if instrumented tests are not feasible) *is determination of field strengths either by direct measurement* or *estimation by analysis of aircraft emitter/antenna characteristics.* Field strengths can then be compared to individual ordnance (or aircraft EID) MAEs to determine potential HERO risks.

In addition, the field strengths can be compared to established personnel limits to determine potential HERP risks for personnel handling/loading ordnance. (Mitigating HERP may have the added benefit of mitigating HERO.) For HERO, the disadvantage of the analytical approach is that simplification (for example, assumption of far field conditions, failure to account for blockages/shadowing, and so forth) may lead to overly conservative (high) estimates of field strengths at ordnance locations. If the field strengths, whether measured or calculated, exceed applicable RADHAZ limits, risk mitigation measures should be identified and evaluated to gauge effectiveness and practical implementation. FIGURE 16 illustrates the approach described above; the preferred order of options is as follows:

a. Intra-system testing.

b. Field strength by measurement.

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c. Field strength estimates based on emitter/antenna characteristics.

Once the environment of the aircraft is determined, minor changes to the ordnance system may not require intra-system testing; the aircraft survey in lieu of intra-system HERO testing is only acceptable for minor changes to ordnance items. A new installation or major modifications to the ordnance system requires instrumented EID testing. Modifications to the aircraft, such as the installation or re-location of aircraft antennas, may also require intra-system HERO testing.

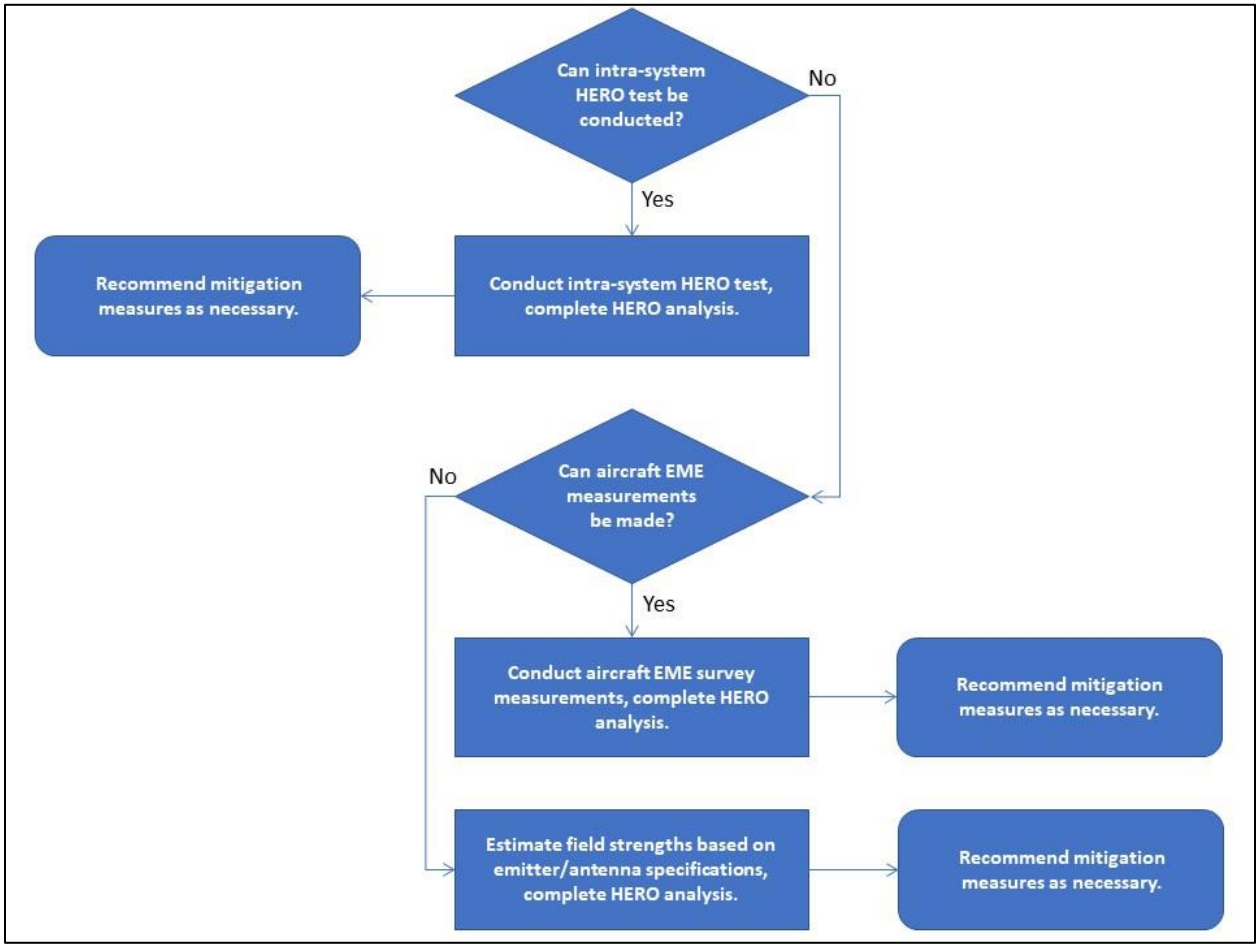


FIGURE 16. Approach to assessing HERO impact of aircraft emitters

4.2.10.4 Test Measurements. As mentioned in the previous section, test measurements can be either an intra-system HERO test involving instrumented EIDs or direct measurement of the E-field strengths produced by the aircraft's emitters. Because intra-system HERO tests are much like "standard" HERO tests at HERO test facilities and are addressed in detail in Part 2 of this handbook, the focus in subsequent paragraphs is on field-strength measurements and field-strength predictions, as opposed to instrumented EID tests. Two types of field strength measurements are discussed: (1) measurements at or close to each ordnance location on the host aircraft, and (2) measurements made at radial distances away from each emitter antenna. The objective of the first type of measurement is to determine the maximum field strengths that can exist at any location at or near the aircraft – platform-loaded, handling/loading locations, and staging locations. The objective of the second measurement type is to determine the reduction in

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fields at increasing separation distances from emitters. This radial data is useful for determining SSDs for HERO SUSCEPTIBLE and HERO UNSAFE ordnance and for determining the distances at which field strengths diminish below specific test-based MAEs.

4.2.10.5 Field-Strength Measurements Near Ordnance Locations. Concern for the effects of emissions at or near specific ordnance locations prompts a concentration of measurements in proximity to those locations. For illustration purposes, FIGURE 17 depicts six hypothetical measurement antenna locations, one for each ordnance item or aircraft EID location.

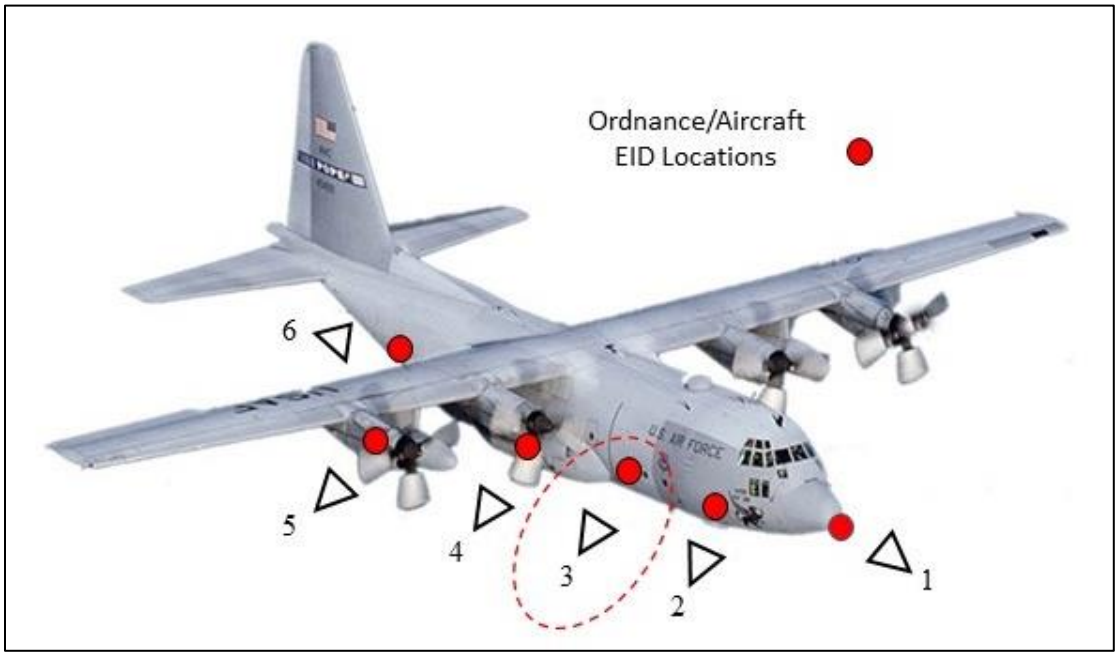


FIGURE 17. Measurement test points near ordnance locations

The objective of taking these measurements is to determine the worst case (maximum) field-strength levels for each aircraft emitter that may pose a HERO risk to the ordnance or around each location.

The measurement antenna should be positioned to maximize pick up, starting very near the attachment point but also including nearby locations (dashed line) where the ordnance item may be handled or staged during ordnance operations. Measurements should be taken at expected ordnance heights as the measurement antenna is directed toward the emitter antenna.

4.2.10.6 Radial Field-Strength Measurements. Field-strength measurements along a radial is a second objective, as a means of determining SSDs for an emitter of interest. These measurements require backing the measurement antenna away from the aircraft along a radial line until the field-strength levels diminish to HERO SUSCEPTIBLE and HERO UNSAFE levels referenced in MIL-HDBK-240-7. SSDs can also be calculated for an emitter of interest using the SSD equations, which is useful for establishing expected separation distances. Determination of SSDs by measurement is particularly important for HF emitters, as the SSDs in this frequency range may exceed 200 feet for HERO SUSCEPTIBLE and 900 feet for HERO UNSAFE ordnance, respectively, for a power output of 500 watts and 1.67 dBi antenna gain.

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FIGURE 18 illustrates a series of measurement locations at progressively increasing distances from the emitter antenna, in this case, an HF wire antenna. This example correlates to ordnance location 5 in FIGURE 17. Each ordnance location should be similarly tested.

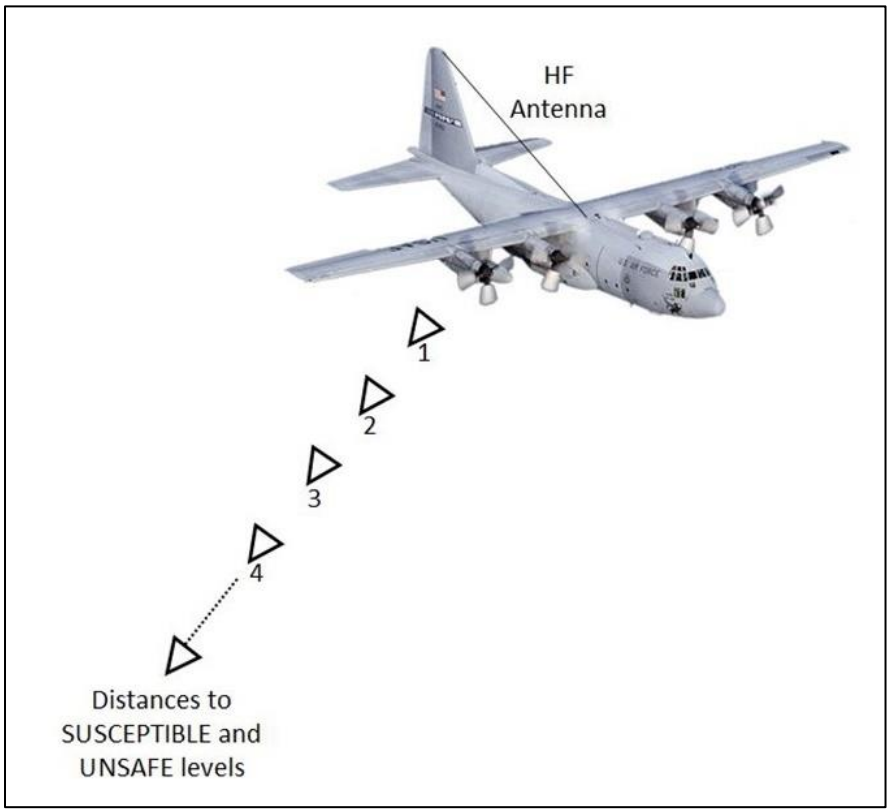


FIGURE 18. Antenna test locations extending radially from the aircraft

4.2.10.7 Field-Strength Predictions. Determination of field strengths by measurement is generally preferred, but if measurements are not feasible, field strengths may be estimated (calculated) based on emitter/antenna characteristics. However, worst-case calculated aircraft-generated EME values may be higher than measured EME values due to losses associated with cabling, antenna directivity, impedance matching, wave propagation/absorption, platform shielding, near-field ordnance locations, and other factors.

The worst-case aircraft-generated EME is calculated by determining the E-field strength produced by the aircraft at the location of each collocated ordnance item using equations (4-8) and (4-7). (These equations assume far-field conditions and direct illumination in the main beam of the transmitting antenna.)

If one or more transmitters operating in the same frequency band are collocated with the aircraft, the effect of the additional transmitters must be accounted for using equation (4-26).

$$E_T = \sqrt{E_1^2 + E_2^2 + E_3^2} \qquad (4\text{-}26)$$

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where:

E_T is the total E-field strength (V/m)
E_1, E_2, E_3 are the individual E-field strengths from other transmitters (V/m)

4.2.10.8 Applying Field-Strength Values. Field-strength values, whether determined by measurement or by analysis, are useful when compared to ordnance sensitivity to assess whether a HERO risk exists and/or provide a basis to mitigate risk. Existence of risk is indicated when field-strength levels exceed applicable MAEs. Thus, wherever field strengths exceed MAEs, a risk exists, and the level of risk is indicated by the difference between incident field strengths and the MAEs for an ordnance item or aircraft EID. Radial measurements are useful in providing the distances at which the field levels diminish below MAEs, a distance known as SSD. SSDs for HERO SUSCEPTIBLE and HERO UNSAFE ordnance categories are the fundamental basis for establishing operational guidance to managing HERO risk.

5. DOCUMENTATION

The purpose of reporting is to present the results of the survey and the conclusions and recommendations drawn from the collected data in a clear and concise manner.

Reporting should document the emitters, ordnance, ordnance areas, ordnance transportation routes, and calculated and measured EMEs. It should discuss any issues that would preclude safe ordnance operations as a result of the operational EME. The information will be used to provide the operational guidance specifically tailored for the platform/installation (see MIL-HDBK-240-7).

Reporting should include, as a minimum, the items listed below:

a. Dates and physical location of survey.

b. Purpose of survey.

c. Mission and ordnance requirements.

d. Emitter systems present.

e. Nomenclature, frequency range, and dynamic range of measuring equipment.

f. Description of testing conditions, include the following:

1) Operational modes of equipment used.

2) Transmitted power of equipment used.

3) Environmental conditions (for example, scaffolding, cranes, and so forth).

4) Measurement locations.

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g. Results, including measured EME levels, locations, and so forth.

h. Conclusions specifying areas of interest.

6. NOTES

6.1 Intended Use. This handbook can be used by any facility engaged in the evaluation of the EME for any of the military departments.

6.2 Supersession Data. Not applicable.

6.3 Cross-Reference of Classification and Substitutability Data. Not applicable.

6.4 Subject Term (Keyword) Listing.

Electromagnetic (EM)
Electromagnetic Environmental Effects (E3)
Electromagnetic Environment (EME)
Electromagnetic Radiation Hazards
Hazards of Electromagnetic Radiation to Ordnance (HERO)
Ordnance

6.5 Identification of Changes. Not applicable.

6.6 Technical Points of Contact. Requests for additional information or assistance on this standard can be obtained from the following:

Air Force

85th Engineering Installation Squadron (EIS/SCYM)
670 Maltby Hall Drive
Building 7701
Keesler AFB, MS 39534
(228) 377-3920
85eis.scym.1@us.af.mil

Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

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Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
Building 8978
Redstone Arsenal, AL 35898-8052
(256) 313-6970
usarmy.redstone.atec.list.rtc-jocg@mail.mil

Army Test and Evaluation Command (ATEC)
White Sands Test Center
Survivability, Vulnerability and Assessment Directorate
ATTN: TEWS-SVE
21225 Headquarters Avenue
WSMR, NM 88002
(575) 678-6307
usarmy.wsmr.atec.list.datts-sv-e@mail.mil

Marine Corps

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Marine Corps Systems Command (MARCORSYSCOM)
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2200 Lester Street
Quantico, VA 22134-6050
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mcsc_safety@usmc.mil

Navy

Commanding Officer
Naval Ordnance Safety and Security Activity
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nossa_hero@us.navy.mil

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Naval Surface Warfare Center
ATTN: B52
5389 Bronson Road, Suite 165
Dahlgren, VA 22448-5153
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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy –SH
Air Force – 11
DISA – DC5

Preparing activity:
DISA (DC5)
(Project EMCS-2022-023)

Review activities:
Army – AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force – 13, 19, 22
DTRA – DS

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